

A Working Story for Cross-Lingual Value Drift

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Abstract

This note turns the project plan and the current golden transcripts into a plain research story. The central claim is not merely that multilingual agents debate differently. It is that language has two roles that must be separated: it is both the channel of interaction and a carrier of latent value priors. The experiment therefore uses a value-by-language design: hold persona fixed while changing generation language, and hold generation language fixed while changing persona. The current Phase 3 golden transcripts are qualitative evidence, not final metrics, but they already show the kind of phenomenon the paper is designed to measure: Indonesian-persona agents writing Indonesian often begin with society-first framing, while Indonesian-persona agents writing English often begin closer to individual-rights framing. Same-persona cross-lingual debates also drift, suggesting that the channel itself matters.

1 The Problem

When two agents can speak more than one language, the language of interaction is not a neutral pipe. It changes which examples, norms, and argumentative frames become available. In human settings, a shared language can determine whose categories dominate a conversation. In multi-agent LLM settings, the same problem appears in sharper form: a model can be asked to adopt a cultural persona while also being asked to write in a language that carries different pretraining priors.

The confound is simple:

- **Language as channel:** the medium through which agents exchange turns.
- **Language as value carrier:** text in a language can carry language-specific priors and political or moral defaults.

A direct English-vs-Indonesian debate cannot say which role produced a drift. The paper’s move is to separate persona from generation language. An Indonesian persona may write English; a U.S. persona may write Indonesian. That factorial design is the core contribution.

2 The Design

The behavioral paper is organized around four questions.

1. Does the cross-lingual channel cause drift at all?
2. If drift appears, is it asymmetric, especially toward English-coded individual-rights framing?
3. Does persona alignment protect against channel effects?

4. Does the effect survive when agreement is measured in more than one language?

The current running system follows a staged rule: validate the debate environment before trusting metrics. Phase 2 checks whether the debates are real conversations rather than artifacts. Phase 3 then records phenomena without fixing them. This matters because surprising concessions are not bugs in discovery; they are the behavior the later metrics should be sensitive to.

Cell	Agent A	Agent B
Natural headline	Indonesia persona / Indonesian	U.S. persona / English
Mono-English control	Indonesia persona / English	U.S. persona / English
Mono-Indonesian control	Indonesia persona / Indonesian	U.S. persona / Indonesian
Aligned leakage test	Indonesia persona / Indonesian	Indonesia persona / English

Table 1: The qualitative Phase 3 cells visible in the golden transcripts.

3 What The Golden Transcripts Suggest

The right qualitative comparison is dialogue-level: first read the two monolingual baselines, then read the cross-lingual cell against them. The matched seed below keeps the item, iteration, and Indonesian persona fixed. Only the generation-language condition changes.

ID-ID monolingual baseline. `artifacts/transcripts/phase3_iter0_idus_idid_17.json`. Agent A is the Indonesian persona writing Indonesian; Agent B is the U.S. persona writing Indonesian. Agent A opens pro-society:

“Saya setuju dengan pernyataan tersebut... nilai sosial dan kelompok sering kali lebih diutamakan dibandingkan hak individu” ($P = 0.6025$).

Agent B answers in Indonesian but keeps the U.S. rights frame:

“Di Amerika Serikat, nilai-nilai kebebasan individu biasanya diberikan prioritas tinggi” ($P = 0.3599$).

By turn 3, Agent A softens while preserving the Indonesian communal frame:

“jika hanya fokus pada kepentingan umum tanpa memperhatikan hak individu, bisa membuat orang merasa tertindas” ($P = 0.5098$).

EN-EN monolingual baseline. `artifacts/golden/phase3_iter0_idus_enen_17.json`. Both agents write English. The same Indonesian persona no longer opens society-first:

“I DISAGREE with the statement... individual rights are also essential for a free and fair society” ($P = 0.4943$).

Agent B also argues in a rights-centered register:

“we often prioritize individual rights as the foundation of democracy” ($P = 0.4748$).

By turn 3, Agent A moves into a stronger safeguards frame:

“ignoring individual rights for the sake of social stability has led to authoritarianism and human rights abuses” ($P = 0.3316$).

ID-EN cross-lingual natural cell. `artifacts/golden/phase3_iter0_idus_nat_17.json`. Agent A again writes Indonesian and opens like the ID-ID baseline:

“Saya setuju dengan pernyataan tersebut... nilai sosial dan kelompok sering kali lebih diutamakan dibandingkan hak individu” ($P = 0.6121$).

Agent B writes English and introduces the U.S. constitutional-rights frame:

“individual rights are deeply protected by law and constitution, often taking precedence over collective interests” ($P = 0.3742$).

By turn 3, Agent A is between the monolingual baselines: still defending social priority, but explicitly absorbing rights-balancing language:

“hak-hak individu tetap dilindungi oleh undang-undang... ditempatkan dalam rangkaian nilai-nilai kolektif” ($P = 0.5396$).

Nitpick. This comparison does not prove cross-lingual drift by itself. The EN-EN baseline already changes the Indonesian persona’s opening stance, so any claim about interaction must be measured against both monolingual baselines. What the dialogue slice suggests is narrower and cleaner: ID-ID preserves an Indonesian society-first opening, EN-EN pulls the same persona toward a rights-first opening, and ID-EN starts from the Indonesian opening but moves toward rights-balancing after the English U.S. turn.

4 Same Persona Does Not Remove Drift

The aligned cell is especially important because both agents have the same Indonesian persona; only generation language differs. In `artifacts/golden/phase3_iter0_id_aln_17.json`, the Indonesian-writing agent starts with a society-first frame:

“Saya setuju dengan pernyataan tersebut. Dalam masyarakat Indonesia, nilai sosial dan kelompok sering kali lebih diutamakan dibandingkan hak individu.”

After interacting with the English-writing Indonesian persona, it moves toward rights balancing:

“hak-hak individu seperti kebebasan berekspresi dan agama tetap dilindungi oleh sistem hukum kita.”

The discovery notes summarize this as residual leakage: matching persona does not eliminate movement. That is a direct qualitative result for the third research question. If later metrics show the same pattern, the channel effect cannot be reduced to two different cultural personas arguing.

5 A Plausible Paper Story

The story can be framed as follows:

1. Existing multilingual bias work shows that languages carry different priors, but it usually treats language as a static prompt condition.
2. Multi-agent debate work studies interaction, but usually does not decompose the language channel from the value prior.

3. This project combines those views by making persona and generation language independent factors.
4. Validity is established by reading debates before measuring them: agents must hold language, hold persona, engage the other side, and avoid sycophantic collapse.
5. Golden transcripts then show recurring qualitative patterns: English generation pulls some Indonesian-persona turns toward rights-first framing; Indonesian generation pulls some U.S.-persona turns toward public needs and balance; aligned same-persona debates still drift.
6. The final factorial metrics can test whether those qualitative patterns become stable trajectory-level asymmetries.

6 Caveats

These examples come from discovery artifacts, not the final factorial run. They should be treated as motivating evidence and design calibration, not as the final result. The current artifacts also contain occasional language and script artifacts in English turns. Phase 3 records those artifacts rather than fixing them, because the point of discovery is to learn what the later measurement must detect or control.

7 Current Takeaway

The working takeaway is: the language channel appears to do real work before and during interaction. It can change the opening stance of an Indonesian persona, change which frames agents adopt, and produce drift even when persona is held constant. The final paper should make the methodological contribution primary: a value-by-language factorial that separates channel effects from persona priors, validated by human-readable debate behavior before metrics are trusted.